Polish Electricity Demand & Forecast Performance

An end-to-end analytics project using Python, PostgreSQL, SQL, and Power BI — analyzing 13,436 observations of system load across 15-minute intervals from January to June 2026.

PYTHON | POSTGRESQL | SQL | POWER BI

Problem Statement

Electricity demand shifts throughout the day and week, creating recurring load patterns. Accurate forecasts are critical for generation planning and maintaining supply-demand balance. Differences between forecasted and actual load reveal periods where demand is harder to predict.

Core Objective

Examine historical demand, evaluate forecast performance, and identify recurring periods of higher demand and larger forecast errors.

Key Questions

- How closely does forecasted load follow actual load?
- When are forecast errors largest?
- How does demand differ weekdays vs. weekends?

Dataset Overview

```
<class 'pandas.DataFrame'>
RangeIndex: 13436 entries, 0 to 13435
Data columns (total 20 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   date                          13436 non-null  str
 1   OREB [Time unit from - to]    13436 non-null  str
 2   forecasted_load_mw            13436 non-null  int64
 3   actual_load_mw                13436 non-null  float64
 4   time                          13436 non-null  str
 5   datetime                      13436 non-null  datetime64[us]
 6   hour                          13436 non-null  int64
 7   weekday                       13436 non-null  str
 8   month                         13436 non-null  int64
 9   Month_Name                    13436 non-null  str
 10  Time_Difference               13435 non-null  str
 11  forecast_error_mw             13436 non-null  float64
 12  absolute_error_mw             13436 non-null  float64
 13  squared_error_mw2             13436 non-null  float64
 14  absolute_percentage_error     13436 non-null  float64
 15  forecast_status               13436 non-null  str
 16  year                          13436 non-null  int32
 17  week                          13436 non-null  int64
 18  weekday_number                13436 non-null  int32
 19  is_weekend                    13436 non-null  bool
dtypes: bool(1), datetime64[us](1), float64(5), int32(2), int64(4), str(7)
```

Source: PSE (Polish Power System Operator). Variables include actual load, forecasted load (MW), time/calendar features, and forecast-error metrics.

Chronological validation identified two irregularities: a one-week gap between reporting periods and a one-hour daylight-saving-time transition — both investigated separately, not treated as missing or duplicated records.

13,436

Observations at 15-minute intervals

20

Final analytical columns

Jan – Jun 2026

24 Jan to 14 Jun 2026

Project Workflow

- **Raw Excel**
- **Python Prep**
- **PostgreSQL**
- **SQL Analysis**

Raw Excel files were loaded and cleaned in Python, stored in PostgreSQL, analyzed via SQL, and visualized in an interactive Power BI dashboard.

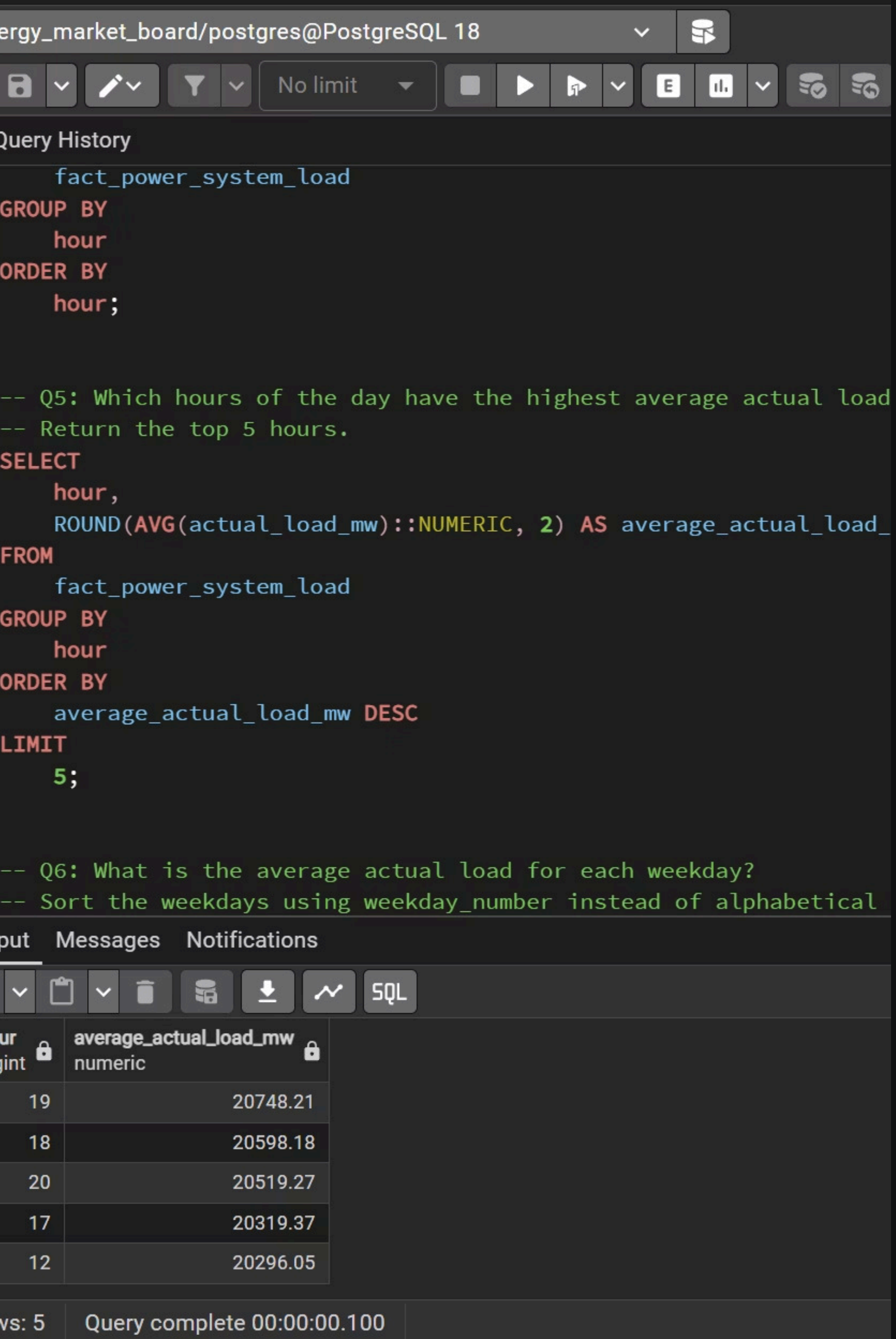
Data Cleaning & Feature Engineering

Data Preparation

- Loaded and concatenated all `.xlsx` files with Pandas
- Removed duplicates, cleaned column names, checked missing values and timestamps
- Created complete datetime field and sorted chronologically

Features Engineered

- **Calendar:** Hour, weekday, week number, month, weekend flag
- **Forecast Error:** Actual − Forecasted (positive = underforecast)
- **Metrics:** Absolute error, APE, squared error
- **Status:** Underforecast / Overforecast / Exact



SQL & Database

The final dataset was loaded into PostgreSQL via SQLAlchemy. SQL was used to calculate aggregations, compare weekdays vs. weekends, identify peak-load periods, and find the largest forecast errors.

```
SELECT
 hour,
 ROUND(AVG(absolute_error_mw)::NUMERIC, 2)
 AS average_absolute_error_mw
FROM fact_power_system_load
GROUP BY hour
ORDER BY average_absolute_error_mw DESC;
```

Key Results

18,498 MW

Average Actual Load

26,787 MW

Maximum Actual Load

491 MW

MAE

2.70%

MAPE

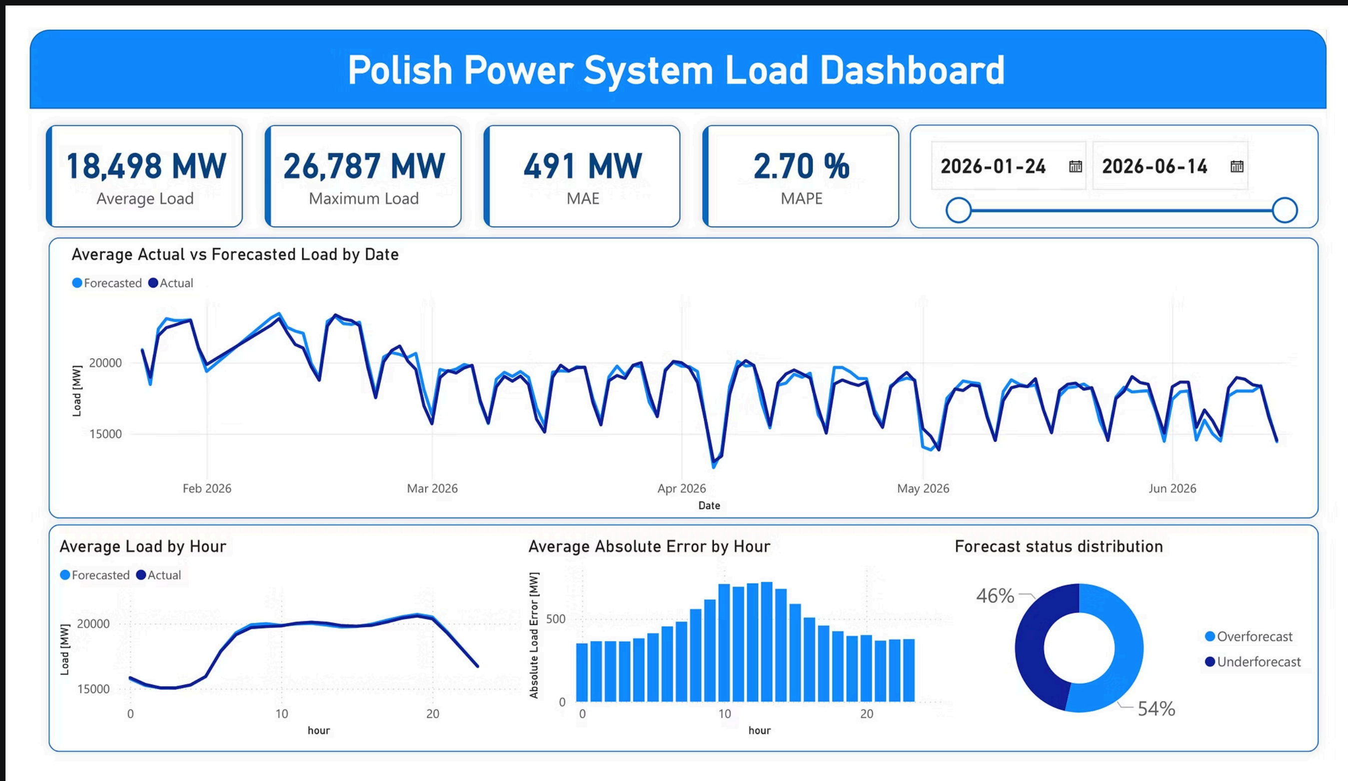
Forecast Bias

Average forecasted load (18,711 MW) was slightly higher than actual. **54.77%** of observations were overforecasts; **45.23%** were underforecasts.

Demand Patterns

Peak average demand occurred around **19:00**. Largest forecast errors occurred around **12:00–13:00** — not during the evening peak. Weekend load (~16,548 MW) was ~**15% lower** than weekday load (~19,511 MW).

Dashboard & Next Steps



Power BI Dashboard

Interactive views of actual vs. forecasted load, average hourly demand, hourly MAE, and forecast-status distribution across the analysis period.

Next Steps

- Integrate electricity price, renewable generation, and cross-border flow data
- Add weather and holiday variables
- Build a short-term load forecasting model
- Automate data ingestion and database updates